Return

Expected stock return is the difference between expected future price and expected average execution price multiplied by the number of shares in the position. Here, the expected average execution price needs to incorporate market impact cost.

For a general market impact function in the absence of natural price appreciation during the trading horizon, the expected execution value for a stock can be computed as follows:

$$\begin{aligned} S_i\overline{P_i} &= \sum_{j=1}^{n} x_{ij}(P_{i0} + MI(x_{ij})) \\ S_i &= \sum_{j=1}^{n} x_{ij} \end{aligned} \tag{10.9}$$

where P_{i0} is the current price, x_{ij} is the number of shares of stock i to trade in period j, and $MI(x_{ij})$ is the market impact cost expressed in \$/share for transacting x_{ij} shares, S_i is the total number of shares, $S_i, x_{ij}, MI(x_{ij}) > 0$ for buys or long positions, and $S_i, x_{ij}, MI(x_{ij}) < 0$ for sells or short positions. Notice that this representation of the market impact cost $MI(x_{ij})$ does indeed provide costs that are dependent upon the underlying trading strategy x.

The total expected dollar return for the stock is:

$$\begin{aligned} \mu_i &= S_iP_{it} - S_i\overline{P}_i \\ &= S_iP_{it} - \sum_{j=1}^{n} x_{ij}(P_{i0} + MI(x_{ij})) \\ &= S_iP_{it} - S_iP_{i0} - \sum_{j=1}^{n} x_{ij}MI(x_{ij}) \end{aligned} \tag{10.10}$$

For an m-stock portfolio, the total expected dollar return accounting for market impact is:

$$\mu_p = \sum_{i=1}^{m} \left(S_iP_{it} - S_iP_{i0} - \sum_{j=1}^{n} x_{ij}MI(x_{ij}) \right) \tag{10.11}$$

Further insight and formulation of market impact models customized for the portfolio construction process can be found at www.KissellResearch.com.

Risk

The risk (variance) of a portfolio over a specified trading period is determined from the number of shares held in the portfolio and the corresponding covariance matrix. The total portfolio risk for either a held portfolio or